

Grade 7
Unit 10
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Using substitution, determine whether the given numerical value is part of the solution set of the inequality. Show your work.

1. Is $p = 12$ part of the solution set for $8p \geq 96$?

$$8(12) \geq 96$$

$$96 \geq 96$$

$p = 12$ is part of the solution set.

2. Is $k = 1$ part of the solution set for $\frac{k}{4} > 1$?

$$\frac{1}{4} > 1 \text{ Not true}$$

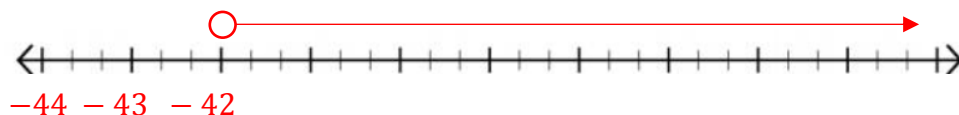
$k = 1$ is NOT part of the solution set.

Use the inequality $\frac{x}{-6} < 7$ to complete the following.

3. Complete the inequality to create an equivalent inequality.

$$x > -42$$

4. Graph the solution set for $\frac{x}{-6} < 7$



5. Find the solution to both inequalities using the multiplication property of inequality.

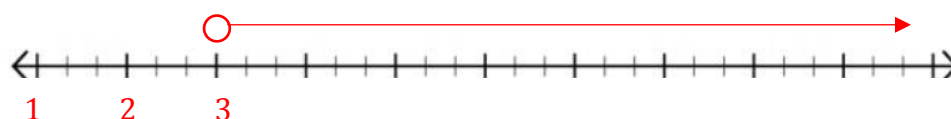
$-4 \cdot -\frac{p}{4} \leq 15 \cdot -4$ $p \geq -60$	$4 \cdot \frac{p}{4} \leq 15 \cdot 4$ $p \leq 60$
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6. Find the solution to both inequalities using the division property of inequality.

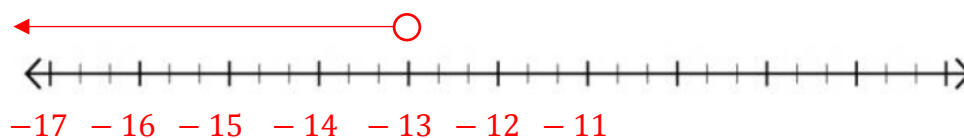
$\frac{7y}{7} < \frac{-105}{7}$ $y < -15$	$\frac{-7y}{-7} < \frac{-105}{-7}$ $y > 15$
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Solve each inequality. Then, graph its solution set.

7. $-2 \cdot -\frac{c}{2} < -1.5 \cdot -2$ $c > 3$



8. $\frac{-4x}{-4} > \frac{52}{-4}$ $x < -13$



9. $\frac{5x}{5} \leq \frac{-55}{5}$ $x \leq -11$



10. How is the process of solving an inequality with a negative coefficient similar to the process of solving an inequality without a negative coefficient?

In both situations, you use the Division property of equality and divide both sides by the negative coefficient.